The algorithm represents the 1-dimensional number line as a binary tree and each node splits the space at a pivot value. When a line segment is inserted, the algorithm recursively places it into the correct regions of the tree. If a line crosses a pivot, it is split and inserted into both subtrees. Nodes are then covered and marked to indicate the entire interval is filled. After all segments are inserted, the total coverage is computed by recursively summing the lengths of covered intervals.

Runtime

Inserting a line: **O(log n)**

n lines insertion: **O(n log n)**

Summing the tree: **O(n)**

Overall expected runtime: O(nlogn)O(n \log n)O(nlogn)